Sports Data Pipeline on AWS · Build Order

Numbered badges show which phase builds each part. Build one phase at a time, one pull request per phase.

Architecture v1.1.0 · 12 Nov 2026 · see docs/architecture/CHANGELOG.md

Planned, not deployed yet

Live / streaming flow

Batch flow

Orchestration trigger / quality check

CI/CD deployment flow

ORCHESTRATION

7 · EventBridge Scheduler
polls on a schedule

schedule

9 · Amazon MWAA (managed Airflow)
or Step Functions for lower cost

DAGs trigger the batch backfill, Glue jobs, then dbt build + tests

SOURCES

7 · Live REST APIs
API-Football · NHL API · MLB StatsAPI · balldontlie · OpenF1

8 · Live feeds
SportScore websockets · OpenF1 live · OpenF1 history (replay)

4 · Historical
StatsBomb Open Data · Jolpica / FastF1 · Cricsheet · nflverse · Kaggle

4 · Reference API
TheSportsDB: teams, players (no live data on free plan)

INGESTION

7 · AWS Lambda pollers
call the REST APIs, free tier ~100 calls/day

8 · ECS Fargate task
websocket listeners + OpenF1 replay simulator

4 · Batch backfill
Lambda or Glue job loads StatsBomb, Jolpica, Cricsheet and reference data into S3 raw

STREAMING

7 · Kinesis Data Streams
live event stream

7 · Secrets Manager
API keys for Lambda, Fargate and backfill

DELIVERY

7 · Kinesis Data Firehose
buffers and writes JSON to S3

8 · OpenF1 note
Live data only during sessions; none between sessions or in the off-season (Dec–Mar). Historical (2023+) is free; real-time may need a paid plan. The replay simulator feeds past sessions in time order to mimic a live stream.

RAW STORAGE

3 · Amazon S3 · raw zone
JSON, partitioned by source / date

PROCESSING

5 · AWS Lambda
Active: stand-in while Glue is blocked
sports-<env>-raw-to-clean
Python 3.13 · arm64 · AWS SDK for pandas layer (pyarrow)
JSON → Parquet at <table>/dt=<date>/
registers tables + partitions in the Catalog
on demand now; orchestrator starts it in phase 9

`processing_engine = "lambda" | "glue"`

5 · AWS Glue job
Planned: waiting on AWS Support
Glue 5.0 · Spark · FLEX · 2× G.1X · same inputs and outputs
Job creation blocked ("Account is denied access"), case open
Deployed: IAM role, log group, code in artifacts bucket.
Missing: the job

CLEAN ZONE

3 · Amazon S3 · clean zone
Parquet tables

3 · Glue Data Catalog
table definitions

WAREHOUSE

6 · Amazon Athena
low cost · or Redshift Serverless

schemas

MODELLING

6 · dbt models
dbt-athena or dbt-redshift
fact_events · dim_team · dim_player · dim_session

OUTPUTS

12 · Amazon QuickSight
dashboards

12 · Notebooks
analysis

DATA QUALITY + MONITORING

10 · Amazon CloudWatch
logs, metrics and alarms for Lambda, Fargate, Kinesis, Firehose, Glue and MWAA

10 · Glue Data Quality
rules on the clean zone

10 · Late or corrected events
keep the latest version per event id; reprocess the affected date partitions

6 · dbt tests
not_null · unique · relationships · freshness

DEPLOYMENT · TERRAFORM + GITHUB CI/CD

2 · GitHub repo
/infra · /ingestion · /dbt · /orchestration · /.github/workflows

2 · GitHub Actions
CI/CD workflows

1 · OIDC → IAM role
assumed by Actions · no stored AWS keys

1 · Terraform
modules in /infra

apply

11 · AWS · dev
full pipeline stack

promote

11 · Manual approval

approved

11 · AWS · prod
full pipeline stack

2 · On pull request
terraform fmt + validate · Checkov / tfsec · plan posted to the PR · pytest for Lambdas · dbt build on dev

11 · On merge to main
Docker image → ECR · package Lambdas · terraform apply to dev · sync DAGs to S3 · run dbt

1 · Terraform state
S3 backend with state locking · separate state for dev and prod

11 · Promote to prod
GitHub Environments approval, then terraform apply to prod

BUILD ORDER · ONE PHASE PER PR

Foundation · Batch path · Live path · Run + harden · Serve

1 · Bootstrap
Terraform state bucket + lock, GitHub OIDC provider, IAM roles (dev, prod)

2 · CI skeleton
PR workflow: fmt / validate, Checkov / tfsec, plan posted to PR, pytest

3 · Storage
S3 raw and clean buckets, Glue Data Catalog database

4 · Batch first
Backfill StatsBomb, Jolpica / FastF1, Cricsheet, TheSportsDB into S3 raw

5 · Processing
Raw JSON → Parquet via Lambda (now) or Glue job (planned, one Terraform switch); register tables in the Catalog

6 · Warehouse + dbt
Athena workgroup; fact_events, dim_team, dim_player, dim_session + tests

7 · Streaming
Kinesis Streams → Firehose → S3 raw; Lambda pollers on EventBridge; Secrets Manager

8 · Fargate
SportScore websocket listener and OpenF1 replay simulator, image in ECR

9 · Orchestration
DAGs / state machine: backfill → Glue → dbt build + test

10 · Quality + monitoring
Glue Data Quality rules, late / corrected events, CloudWatch alarms

11 · Deploy pipeline
Merge-to-main workflow; prod promotion with GitHub Environments approval

12 · Outputs
QuickSight dashboards, analysis notebooks